1. Introduction to Machine Learning:

- EU AI Act and risk levels
- Definitions of AI, ML, and Deep Learning
- ML approach vs. traditional approach
- Feature extraction

2. Supervised Learning (definition and model evaluation)

- Unsupervised, Supervised, Reinforcement Learning
- MAE, MSE and RMSE
- ROC Analysis
- Dichotomous classifier and Confusion matrix
- Sensitivity, Specificity, Precision, Accuracy and F1-score
- ROC Plane, ROC Curve and AUC

3. Supervised Learning (generalization)

- Linear regression without bias
- Linear regression with bias
- Underfitting and Overfitting
- Model Capacity and Hypothesis Space
- Principle of Parsimony - Occam's Razor

4. Regularization and Hyperparameter Optimization

- No Free Lunch Theorem
- Weight Decay and Hyperparameters
- Regularization
- Hold-out Validation
- Nested Hold-out Validation
- Stratified Hold-out and Stratified Nested Hold-out

5. Cross-Validation and Nested Cross-Validation

- K-Fold Cross-Validation and LOOCV
- Nested K-Fold Cross-Validation

6. Clustering

- Compactness-based Clustering: K-Means, K-Means++
- Model-based Clustering: Gaussian Mixture Model
- Density-based Clustering: DBSCAN, Density Peaks
- Self-Organizing Maps
- Choosing the number of clusters: Elbow method, Silhouette Score, AIC, BIC
- Adjusted Rand Index

7. Dimensionality Reduction

- PCA
- Linear Discriminant Analysis
- Multi-Dimensional Scaling
- IsoMaps
- t-SNE
- UMAP
- AutoEncoders and Variational AutoEncoders

8. Regression/Classification with Linear Models

- Linear and Logistic Regression
- Regularization: Ridge L2, Lasso L1, Elastic net L1+L2, Firth's logistic regression
- Multi-class Classification (One-vs-All, One-vs-One) and Softmax Regression
- Entropy and Cross-Entropy

9. Decision Trees and Ensemble Methods

- CART
- Cost Complexity Pruning
- Bagging
- Random Forest
- ADABOOST
- Gradient Boosting

10. Support Vector Machines

- Large Margin Classifiers
- Kernel Trick and Mercer's theorem
- Soft Margin Classifier

11. Explainability

- LIME
- SHAP